MIL-E-1/1531A(NAVY)
22 November 1974
SUPERSEDING
MIL-E-1/1531(NAVY)
1 May 1964
(See note 38)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRAVELING WAVE

TYPE M5058A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Low-noise amplifier, 4.3 to 7.35 GHz, periodic permanent magnet focused

ABSOLUTE RATINGS:

Parameter:	Ef	If	Ec1	Ic1, 2, 3, 4, and 5	Ec2	Ec3	Ec4, 5	Ew	Iw
Unit:	V	A	Vdc	uAdc	Vdc	Vdc	Vdc	Vdc	uAdc
Maximum:	6.6	0.4	-150	100	100	500	1,000	1,500	500
Minimum:	6.0	0.1	---	---	-150	---	---	---	---
			(Notes 11, 25)		(Notes 11, 25, 32)	(Notes 11, 25)	(Notes 11, 25)	(Notes 11, 25)	

ABSOLUTE RATINGS:

Parameter:	Eb	Ib	Ik	Pi	TA	tk	If (surge)	Alt
Unit:	Vdc	mAdc	mAdc	dBm	°C	sec	a	ft
Maximum:	1,500	1.0	1.0	40	100	---	1.0	80,000
Minimum	---	---	---	---	-65	60	---	---
	(Notes 11, 25)		(Notes 11, 25)		(Note 28)			

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Connectors: See figure 1
Cooling: Convection
Coupling: Coaxial
Marking: See note 2
Mounting position: Any
Weight: 5 pounds maximum (2.27 kg)

TEST CONDITIONS:

Parameter:	Ef	If	Ec1, 2, 3, 4, & 5	Ew	Eb	Ik	Pi
Unit:	V	A	Vdc	Vdc	Vdc	mAdc	dBm
6.3	Setup from data on tube label and individual test data sheet per note 2.						

Frequency	
F1	4.3 GHz
F2	5.0 GHz
F3	5.8 GHz
F4	6.5 GHz
F5	7.35 GHz

GENERAL:

First article approval - Required (see note 37)
Shelf life - 12 months
Preparation of data - See note 30
Packaging - See note 24

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	First article testing	37					
1011	Humidity	13, 21	No voltages	---	---	---	---
1006	Salt spray	13, 21	No voltages	---	---	---	---
1136	Rough handling	23		---	---	---	---
---	Vibration	13, 16, 35	F = 5 to 55 Hz; 0.06 inch double amplitude	ΔGss	---	0.5	dB
---	Shock	5, 13, 16, 35		ΔGss	---	0.5	dB
	Quality conformance inspection, part 1	6					
4289	Heater current	12		If	0.19	0.24	A
1266	Grid current	12		Ic1-Ic5	0	50	μAdc
4253	Gain, small signal	7, 17	Po = 0 dBm (max)	Gss	35	48	dB
4253	Power gain	8, 9	F = F1 thru F5	G	---	---	dB
4251	Output power and gain variation	7, 17 7, 17	Small signal Fine grain gain characteristics	ΔGss ---	--- ---	4 ±0.75	dB dB
4260	Noise figure	29	F = F1 thru F5	NF	---	10	dB
4254	Saturated rf power output	3	F = F1 thru F5	Po	10	20	dBm
4256	Input and output match	1	No voltages	VSWR	---	2.0:1	---
---	Overdrive power	19	F = F1 thru F5	---	---	---	---
---	Gain compression	31	F = F1 thru F5	ΔGss	---	1	dB
	Quality conformance inspection, part 2						
4261	Stability	26	No rf input	---	---	---	---
4257	Amplifier insertion loss	10	Pi = 10 dBm; no voltages	L1	70	---	dB
4261	Spurious output: Input terminal Output terminal Output modulation	 36 36 33	 No rf input No rf input F = F3	 --- --- ---	 --- --- ---	 -65 -30 ---	 dBm dBm ---
4263	Gain, control (small signal)	32		ΔGss	70	---	dB
4266	Direct-interelectrode capacitance	4	No voltages	C	---	40	pF
4258	Radio interference shielding	34	F = F1, F3, F5	---	---	---	---
1357	Magnet polarity and strength	18		---	---	---	---
---	Temperature, altitude	14, 15, 20		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	22, 27	Group D: F = F1, F3, F5	---	1,000	---	Hours
---	Life-test end points:						
4253	Gain, small signal	7, 17	Po = 0 dBm (max)	Gss	35	48	dB
4260	Noise figure	29	F = F1 thru F5	NF	---	10	dB
---	Overdrive power	19	F = F1 thru F5	---	---	---	---

NOTES:

1. The VSWR shall be measured across the frequency band using a swept frequency technique.
2. The tube operating voltages with respect to the cathode, and currents, shall be listed on a label affixed to the tube. Voltages shall fall within the following limits:

Element	Minimum voltage	Maximum voltage
Grid 1	0	-50
Grid 2	0	50
Grid 3	0	100
Grid 4	50	250
Grid 5	175	500
Helix	700	875
Collector	875	955

   The cathode, helix or collector may be at capsule potential.
3. The saturation power output test shall be performed as follows: At each frequency, increase the power input until the first power output peak is reached. Measure the power output at this value of power input.
4. Measure the capacitance between grid 2 and all other leads connected together, including capsule.
5. Subject the operating tube at prevailing room temperature to 18 half sine wave impact shocks of 15 G and a time duration of 11 ±1 ms. Three impact shocks shall be applied in each direction to the tube in each of three mutually perpendicular axes. After completion of the test, the tube shall meet the limits of the gain, saturated power output, and noise figure tests at F3.
6. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
7. Gain tests shall be performed with input power adjusted for output power of 1 milliwatt or less.
8. Power gain is defined as the gain as measured with the output power adjusted to give saturation power output as specified in note 3.
9. Power gain shall be within 8 dB of small signal gain.
10. The isolation shall be measured across the frequency band using a swept frequency technique.
11. Installation and alignment shall be as follows:
    (a) Connect power supply and rf lines to the TWT.
    (b) Apply heater voltage for a period of 1 minute.
    (c) Set grid 2 to zero volts, and set all other elements to voltages shown on the tube label.
    (d) Turn up the voltage on grid 2 until the voltage reaches the value shown on the tube label. The cathode current shall be approximately that indicated on the label.
12. At any temperature between -54° and +95°C, and with the rf input power adjusted to give saturated power output (see note 3), the current(s) shall not exceed the specified limits(s).
13. Follow procedure of MIL-T-5422.
14. Conditions shall be in accordance with MIL-T-5422, "Test Chamber conditions for temperature-altitude test" table, class 2.

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NOTES: -Continued

15. Performance at the operating temperature extremes shall be compared to the performance at 25°C and shall not exceed the following limits.

Performance	Limits
ΔGain	±2.0 dB
ΔSaturation power output	±1 dB
ΔNoise figure	±1 dB

16. Measure gain using a swept frequency technique. Gain variation during test shall be less than specified at any frequency between F1 and F5.

17. A swept frequency signal source shall be employed.

18. The performance shall not be affected by magnetic fields with a strength of 30 gauss.

19. Overdrive power shall be 0 dBm for input signals up to and including -7 dBm; and +5 dBm for input signals up to and including -10 dBm.

20. The temperature altitude test shall be performed as follows: Affix a temperature indicating device to the capsule outside diameter at a point approximately bisecting the tube length. Place the tube in chamber and adjust chamber temperature to -62°C. Maintain this condition for 1 hour. Adjust the chamber temperature to -54°C. When the capsule temperature has stabilized, apply the voltages and measure the performance. Proceed with the temperature altitude test as specified herein. The tube shall be operating during all steps and where a test is required (end of steps 1, 3, and 5) perform the specified tests only after the capsule temperature has stabilized.

Step	Condition	Time	Test
1	T = 25°C, Alt = 0 ft	1 hour	Required
2	T = 25°C, Alt = 70,000 ft	10 minutes	Not required
3	T = 71°C, Alt = 0 ft	1 hour	Required
4	T = 71°C, Alt = 45,000 ft	10 minutes	Not required
5	T = 95°C, Alt = 0 ft	30 minutes	Required

21. After completion of the test, the tube shall meet the limits of the gain, saturated power output, and noise figure tests.

22. At prevailing room ambient temperature, and with the rf input power adjusted to give saturated power output (see note 3), and the grid and helix currents shall not exceed the specified limits.

23. One tube shall be packaged and packed in its regular shipping container. The packaged tube shall then be dropped from a height of 30 inches on to a rigid horizontal surface four times; once each on the top and bottom and once each on two adjacent sides of the container. Following this test, the tube shall meet the limits of the gain, saturation power output, and noise figure tests at F3.

24. Preservation, packaging, and packing. Unless otherwise specified in the contract or order, preservation, packing, and packing shall be as follows:

(a) The package shall be so constructed that the tube can be removed for inspection and repacked.
(b) The package shall be capable of 1 year storage under conditions as specified in MIL-E-5400.

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NOTES: -Continued

25. In order to maintain the specified performance over the specified temperature range, the following power supply requirements shall be met:

Element	Installation accuracy (percent)	Regulation (percent)
Heater	1.0	1.0
Grid 1	1.0	1.0
Grid 2	1.0	1.0
Grid 3	1.0	1.0
Grid 4	1.0	1.0
Helix	1.0	1.0
Collector	5.0	5.0

(a) Installation accuracy is set on accuracy at 25°C.
(b) Regulation includes power supply variations from all causes including temperature.

26. With the tube input terminated in a short, the phase of the mismatch shall be varied by 360 electrical degrees and helix voltage swept plus and minus 5 percent from its optimum value, at a 60 to 400 Hz rate. The detected tube output shall be viewed as vertical deflection on an oscilloscope and horizontal deflection as proportional to helix voltage. No oscillations, as indicated by discontinuities in the oscilloscope trace, shall be evident. This test shall then be repeated with the input and output of the tube reversed.

27. Before the life tests and at 50, 100, 200, 500, and 750 hours, the gain, saturation power output, and noise figure shall be measured. At the end of the life tests, the tube shall meet the limits of the gain saturated power output, and noise figure at 25°C.

28. The tube shall be capable of satisfying the requirements of this TSS at the specified ambient temperatures.

29. For noise figure measurements, see figure 2.

30. One reproducible and two prints of quality conformance inspection data shall be supplied to the procuring activity.

31. The tube shall exhibit no more than the specified gain compression at the 3 milliwatt power output level.

32. The tube shall exhibit a minimum of 70 dB ΔGss when the grid 2 voltage is at -50 volts dc.

33. All spurious signals shall be at least 60 dB below the carrier.

34. With the tube immersed in a uniform field of 1W/m^2 of arbitrary polarization and with the input terminal terminated into a matched load, the signal appearing at the output shall not exceed -20 dBm.

35. After this test, the tube shall meet the specified rf performance at F3.

36. To calibrate and measure the input terminal spurious output (see figure 3), the following procedure shall be observed:

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NOTES: -Continued

Step	Action	Item	Operation
1	Connect	Thermistor	To the oscillator output.
2	Operate	Oscillator	CW at F3.
3	Set	Oscillator and variable attenuators	To read zero dBm on the power bridge.
4	Disconnect	Thermistor	From the oscillator output.
5	Add	75 dB of attenuation	Using items 2 and 3 of figure 3.
6	Connect	Oscillator output through 75 dB of attenuation	To the test equipment low-noise amplifier.
7	Set	Oscilloscope	To the 0.85 MHz bandwidth position, and 1 μ/sec rate.
8	Adjust	Gain of low-noise TWT test equipment amplifier at F3	For a readable deflection on the oscilloscope face.
9	Mark	Midline	Of the oscilloscope deflection.
10	Connect	Input terminal of the tube under test	The straight through arm of the 10 dB directional coupler attached to the sliding short.
11	Attach	Test equipment low-noise amplifier input	To the 10 dB down arm of the directional coupler.
12	Observe	Scope deflection.	As a sliding short is moved through 360° phase variation and monitored through a 10 dB coupler on the input terminal. The presentation of the oscilloscope spurious output shall not exceed the mark of the oscilloscope face recorded in step 9.

To calibrate and measure the output terminal spurious output, repeat the above procedure, except:

(a) In step 5 change 75 dB to 40 dB, and
(b) In step 10 connect the output terminal of the tube under test to the straight through arm of the 10 dB directional coupler.

37. First article inspection shall consist of performing all tests listed on this tube specification sheet. The sample size and allowable defects shall be in accordance with MIL-E-1, Qualification, samples, and acceptance criteria. Three copies of the test report, prepared in accordance with MIL-STD-831, shall be forwarded to the purchasing activity for evaluation by the preparing activity.

38. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

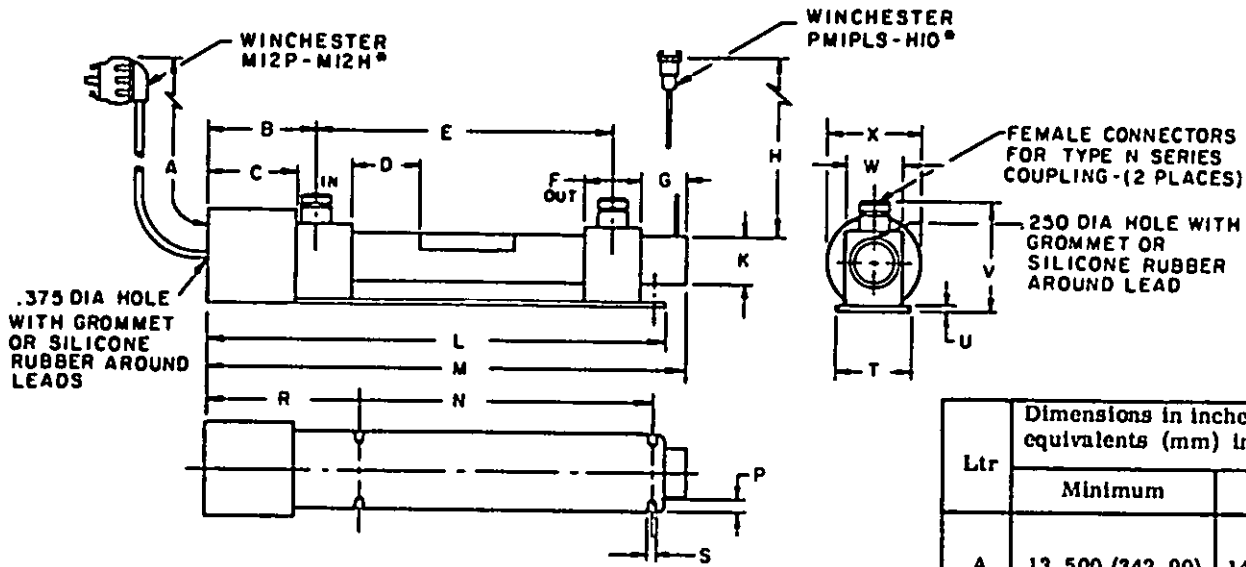
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

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Power plug connections M12P - M12H*

A.	N.C.	-	M. Grid 5	Green/black trace
B.	Grid 1	Green	N. Grid 4 (anode 3)	White
C.	N.C.	-	PM1PLS-H10* collector	Red AWG #20
D.	Capsule ground	Black		
E.	Grid 2 (anode 1)	Blue		
F.	Heater cathode	Yellow		
H.	Heater	Brown		
J.	Cathode	Yellow		
K.	Grid 3 (anode 2)	Gray		
L.	Helix	Orange		

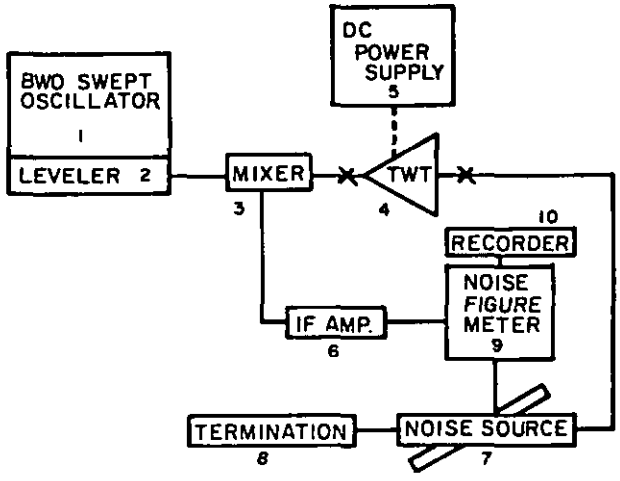
*Or equivalents (If an equivalent is used, specific approval shall be obtained from the bureau or agency concerned prior to its use.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	13.500 (342.90)	14.00 (355.60)
B	2.844 (72.23)	2.906 (73.82)
C	2.375 (60.33)	2.438 (61.92)
D	2.563 (65.09)	2.813 (71.44)
E	9.344 (237.33)	9.406 (238.92)
F	1.469 (37.30)	1.531 (38.90)
G	1.219 (30.95)	1.281 (32.55)
H	6.75 (171.45)	7.25 (184.15)
K	1.219 (30.95)	1.281 (32.55)
L	13.625 (346.08)	13.875 (352.43)
M	14.125 (358.78)	14.375 (365.13)
N	4.109 (104.38)	4.141 (105.17)
P	.302 (7.67)	.322 (8.18)
R	4.109 (104.38)	4.141 (105.17)
S	.178 (4.52)	.198 (5.03)
T	1.984 (50.40)	2.016 (51.20)
U	.109 (2.78)	.141 (3.57)
V	2.813 (71.44)	2.938 (74.61)
W	1.469 (37.30)	1.531 (38.90)
X	2.484 (63.10)	2.516 (63.90)

FIGURE 1. Outline drawing of electron tube type M5058A.

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A. Equipment layout.



B. Test equipment.

1. Microwave oscillator
2. Microwave leveler (*optional)
3. 10 dB directional coupler
4. Test item
5. TWT power supply
6. IF amplifier
7. Noise source
8. Termination
9. Noise figure meter
10. Recorder.

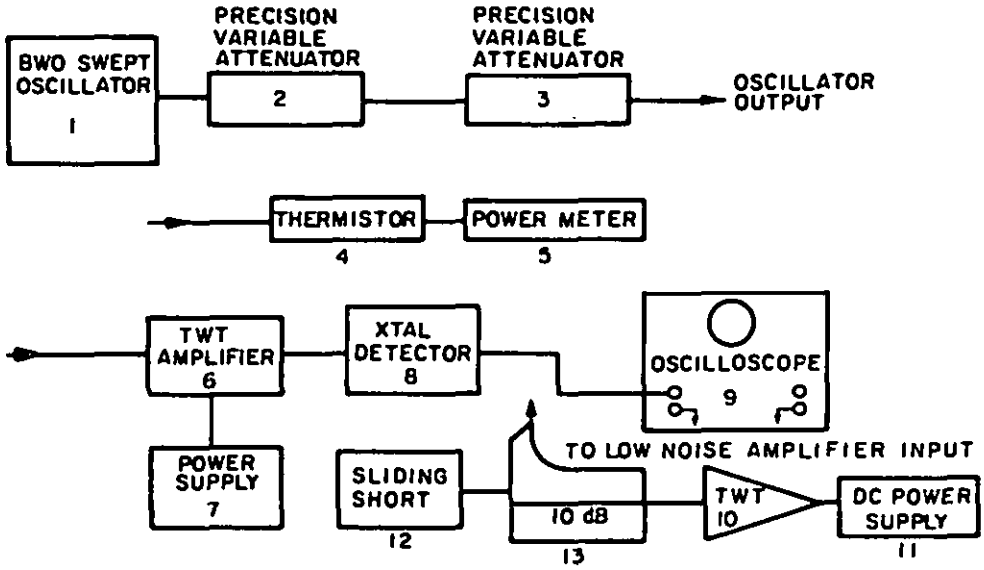
C. Test procedure.

Set oscillator for CW and, operating at fixed frequencies, record noise figure as read from noise figure meter. As an alternate, sweep frequency noise figure measurement technique may be employed.

FIGURE 2. Noise figure test and measurement.

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A. Equipment layout.



B. Test equipment. The following test equipment (or equivalents) shall be used. If an equivalent piece of test equipment is used, specific approval shall be obtained from the bureau or agency concerned prior to its use.

1. Microwave oscillator.
2. Variable attenuator.
3. Variable attenuator.
4. Thermistor mount.
5. Microwave power meter.
6. TWT amplifier, 15 dB maximum. Noise figure.
7. TWT amplifier power supply.
8. Crystal detector.
9. Oscilloscope, Tektronix 531A, with type-D plug-in unit.
10. Tube under test (with directional coupler and sliding short).
11. TWT power supply.
12. Sliding short.
13. 10 dB directional coupler.

FIGURE 3. Spurious output test and measurement.

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